

Table of Content

1.0 Introduction.....	2
1.1 Background of Case Study.....	2
1.2 Problem Statement.....	2
1.3 Objective.....	2
1.4 SDG.....	3
1.5 Project KPI.....	3
2.0 Review on Existing Solution.....	4
2.1 Commercial Solutions (Budgeting Apps & Templates).....	4
2.2 Regulatory Policies & Movements.....	5
2.3 Open-Source & Developer Solutions.....	6
2.4 Proposed Solution.....	7
3.0 Methodology.....	8
3.1 Design Thinking and Planning.....	8
3.2 User Research and Problem Validation.....	9
3.3 Interface Flow Design and UI/UX Prototyping.....	10
4.0 Results.....	11
4.1 Centralized Financial Tracking.....	11
4.2 Visualization of Behavioral Intervention.....	11
4.3 Simulated Automated Invoice Analysis.....	11
4.4 Prototype Evaluation and Observations.....	12
5.0 Prototype [Output & UI Screenshot].....	13
6.0 AI Acknowledgement.....	15
7.0 Conclusion.....	16
References.....	17

1.0 Introduction

1.1 Background of Case Study

The use of "Buy Now, Pay Later" (BNPL) services, such as SPayLater, has grown rapidly in recent years, becoming a popular choice among online shoppers (Maggio et al., 2022). These platforms offer significant convenience, allowing users to make purchases without prior credit history or even formal employment. This makes them particularly accessible to university students and young adults managing limited budgets (Shah, 2025). This ease of access broadens financial inclusion and expands financial opportunities for those who might otherwise be excluded.

1.2 Problem Statement

While BNPL services lower the barrier to entry for consumer goods, users frequently overspend without fully understanding the long-term financial impact of their transactions. Specifically, consumers struggle with the following core issues:

1. **Fragmented Debt:** Managing and visualizing multiple BNPL plans simultaneously across different platforms.
2. **Opaque Obligations:** Tracking the total accumulation of future payment obligations.
3. **Complex Terms:** Understanding hidden fees, compounding interest, and penalties buried in terms of service.
4. **Impulse Purchasing:** Making informed, rational decisions about using BNPL versus alternative payment methods at the point of sale.

1.3 Objective

To address the challenges posed by BNPL services, this project introduces **Bijak Pay**, an application that aims to:

1. **Centralize Financial Tracking:** Develop an automated dashboard that aggregates all active BNPL debts and calculates precise upcoming payment totals.
2. **Promote Behavioral Interventions:** Implement a proactive “nudge” system that alerts users to the financial burden of a new BNPL purchase immediately upon invoice generation, prompting reconsideration or refunds.
3. **Automate Term Analysis:** Utilize AI to automatically extract, analyze, and simplify interest rates, payment terms, and hidden fees from uploaded invoices, empowering users to extract meaningful information and make informed financial decisions.

1.4 SDG

The project's objectives **align with** the United Nations Sustainable Development Goals:

1. **SDG 4 – Quality Education:** By enhancing financial literacy, Bijak Pay empowers users to understand their financial standing, hidden fees, and the long-term impact of their spending habits.
2. **SDG 11 – Sustainable Cities & Communities:** By fostering community-centered tools that promote financial health, the project helps protect users from predatory debt cycles and encourages responsible financial behavior within digital communities.

1.5 Project KPI

The success of Bijak Pay will be measured against the following targets:"

1. **AI Extraction Accuracy:** Achieve a >90% accuracy rate in using AI to extract the correct interest percentages and repayment months from raw invoice data.
2. **Intervention Success Rate:** Measure the percentage of simulated users who opt to refund or cancel a BNPL purchase after receiving the AI-generated rational analysis nudge.
3. **Processing Latency:** Ensure the automated dashboard updates and delivers the behavioral nudge within seconds of a successful BNPL payment receipt.

2.0 Review on Existing Solution

The current landscape of Buy Now, Pay Later (BNPL) management includes a range of solutions developed by industry, policymakers, and the developer community. These solutions can be broadly categorized into commercial financial tracking applications, regulatory policy initiatives, and open-source financial management tools.

2.1 Commercial Solutions (Budgeting Apps & Templates)

The market is saturated with personal finance applications such as YNAB, Copilot, and Monarch Money, alongside manual tracking templates commonly sold on digital marketplaces. These tools are designed to help individuals monitor spending, track financial goals, and manage personal budgets. Many premium budgeting apps offer features such as bank account synchronization and automatic expense categorization, enabling users to track transactions across multiple financial accounts in real time (NerdWallet, 2025; PersonalFinances, 2026).

- **The Good:** These solutions are effective for macro-level budgeting. Premium applications provide automated expense tracking and financial dashboards, while manual spreadsheets or templates allow users to actively engage with budgeting strategies and debt repayment planning.
- **The Limitations:** However, these solutions are largely reactive. They typically present financial data only after a transaction has already occurred. Most budgeting apps primarily analyze transaction-level data from bank feeds rather than extracting detailed financial terms from invoices, which limits their ability to interpret hidden fees, compounding interest structures, or the full repayment schedule associated with services such as Buy Now, Pay Later.

2.2 Regulatory Policies & Movements

Governments around the world are increasingly recognizing the financial risks associated with Buy Now, Pay Later (BNPL) services and have begun introducing regulatory frameworks to enhance consumer protection. In the United Kingdom, the Financial Conduct Authority (FCA) has announced that BNPL products will come under formal regulation beginning in July 2026. Under these rules, lenders will be required to conduct affordability checks, provide clear repayment information, and offer support for consumers experiencing financial difficulty (FCA, 2026a; FCA, 2026b).

Similarly, in the United States, the Consumer Financial Protection Bureau (CFPB) has examined BNPL products under the Truth in Lending Act and Regulation Z, classifying certain BNPL digital accounts as credit devices subject to consumer disclosure and dispute-resolution requirements (CFPB, 2024).

Australia has also moved toward stricter regulation by requiring BNPL providers to obtain credit licenses and perform financial suitability checks to reduce the risk of consumer over-indebtedness (News.com.au, 2025).

- **The Good:** These regulatory movements aim to increase transparency, improve consumer protection, and ensure lenders perform responsible affordability checks before approving BNPL purchases. By introducing disclosure requirements and consumer protections similar to traditional credit products, regulators attempt to reduce the likelihood of predatory lending and excessive debt accumulation.
- **The Limitations (The Gap):** However, regulatory changes typically take years to implement and focus primarily on ensuring lender compliance rather than influencing consumer decision-making. While such policies improve transparency and fairness in lending practices, they do not directly address behavioral factors such as Fear of Missing Out (FOMO) or present bias, which often drive impulsive BNPL purchases.

2.3 Open-Source & Developer Solutions

In addition to commercial applications, developers have attempted to address personal finance tracking challenges through open-source software projects. Several community-driven financial management platforms are available on repositories such as GitHub, where developers publish customizable expense trackers, budgeting dashboards, and personal finance management systems. Examples include open-source tools such as **Firefly III**, a self-hosted personal finance manager designed to track expenses, budgets, and financial accounts, and **Money Manager Ex**, a cross-platform financial management application for monitoring personal cash flow and budgeting. These platforms demonstrate how open-source communities attempt to provide transparent and customizable alternatives to proprietary financial software (Firefly III, 2025; Money Manager Ex, 2025).

Additionally, various experimental developer projects hosted on GitHub implement financial dashboards using frameworks such as Python, Flask, and JavaScript. For example, open-source finance managers built using Flask and MongoDB provide features including expense tracking, budgeting analytics, and reporting dashboards for personal financial monitoring. These projects highlight how developers experiment with automated financial aggregation and visualization tools through self-hosted applications and modular software architectures (GitHub Finance Manager Project, 2025).

- **The Good:** Open-source financial tools offer significant advantages such as transparency, customization, and cost-free accessibility. Because their source code is publicly available, developers can extend functionality, automate financial data processing, and adapt the tools to suit specific financial tracking requirements.
- **The Limitations:** However, these solutions typically require technical expertise to install and maintain. Many open-source finance tools must be deployed through self-hosted environments or containerized systems such as Docker, which requires users to configure servers, databases, and programming environments. Consequently, while these solutions are powerful for technically skilled users, they remain largely inaccessible to the average consumer seeking simple and intuitive financial management tools.

2.4 Proposed Solution

To address the limitations of existing BNPL financial management tools—such as reactive tracking, limited regulatory influence on consumer behavior, and technically complex open-source solutions—this project proposes a proactive BNPL financial awareness system.

The system integrates automated financial tracking with behavioral interventions through three core features:

- **Proactive Behavioral Nudges:** AI-generated notifications are delivered immediately after BNPL transactions, targeting the post-purchase window when users are most likely to reconsider impulsive decisions.
- **Automated Invoice Analysis:** AI-assisted parsing extracts repayment schedules, interest rates, and hidden fees from uploaded BNPL invoices, providing users with a clear view of long-term financial obligations.
- **Accessible Automated Tracking:** Advanced financial tracking is delivered via a simple, user-friendly mobile interface, removing the technical barriers of traditional tools.

By combining real-time financial insights with behavioral interventions, the system bridges the gap between transparency and informed decision-making, promoting more responsible BNPL usage.

3.0 Methodology

To develop a solution that addresses both the behavioral and practical challenges of Buy Now, Pay Later (BNPL) financial management, a user-centered design methodology was adopted. The team employed an agile, rapid design approach combining elements of Design Thinking, problem validation, and UI/UX prototyping to efficiently build and test the proposed solution.

3.1 Design Thinking and Planning

The project followed the Design Thinking framework to guide the ideation and development process. The five core stages—Empathize, Define, Ideate, Prototype, and Test—were utilized to structure a lean and highly focused development cycle.

Empathize and Define

During the initial stage, the team analyzed common financial challenges associated with BNPL services. Particular attention was given to the behavioral gap between the moment of purchase and the long-term financial consequences of installment payments. This stage focused on identifying the psychological disconnect—often referred to as "present bias" or immediate gratification—that leads to impulsive purchasing decisions.

Ideation

Following problem identification, brainstorming sessions were conducted to explore potential interventions that could help users make more informed financial decisions. This process led to the concept of a post-purchase behavioral nudge, where Bijak Pay would present users with a simplified, data-driven explanation of their financial commitment immediately after a BNPL transaction occurs.

Project Coordination

To ensure streamlined project coordination and agile task management, planning and collaboration were centralized through Notion. The platform was utilized to build structured databases for organizing research notes, Kanban boards for assigning tasks among team members, and timelines to track development progress throughout the design process.

3.2 User Research and Problem Validation

Before designing the interface, the team conducted preliminary secondary research to validate the relevance of the identified problem and better understand user behavior related to BNPL services.

Behavioral Analysis

Existing reports and desk research on BNPL usage were reviewed to identify patterns of overspending, late payment penalties, and the demographics most likely to rely on installment-based payment services. This analysis provided insight into the specific financial risks associated with fragmented debt.

Persona Development

Based on the research findings, user personas were developed to represent the primary target audience. The primary persona, "The Aspirational Student," represents university students and young adults who frequently shop online, are highly tech-savvy, but lack structured financial tracking habits. This persona often relies on BNPL to manage limited monthly allowances, making them highly susceptible to accumulating hidden debt across multiple platforms. Establishing this profile helped guide the design decisions for Bijak Pay's interface and feature prioritization.

Identifying the Behavioral Intervention Window

The research indicated that the period immediately after a purchase confirmation represents a critical "cooling-off" moment when users may still reconsider their financial decision. This insight directly informed the design of Bijak Pay's core feature—a notification-based intervention designed to intercept the user and prompt reflection on the true financial implications of their purchase.

3.3 Interface Flow Design and UI/UX Prototyping

After defining the problem and identifying the target users, the team proceeded to design the application structure and user interaction flows. A high-fidelity prototype was developed using Figma to simulate the proposed solution.

The overall user journey was mapped to illustrate how users interact with Bijak Pay from onboarding to active financial monitoring. The primary UX goal was to reduce cognitive load—minimizing user friction while maximizing the visibility of complex financial obligations.

Three primary interaction flows were designed:

1. Aggregated Dashboard Flow

The dashboard serves as the main interface where users can view a consolidated summary of all active BNPL commitments. The design utilizes strict visual hierarchy, instantly drawing the user's eye to the total outstanding balance and upcoming payment deadlines.

2. AI Invoice Scanner Flow

This flow demonstrates how users upload a BNPL receipt or invoice. The interface then displays how Bijak Pay extracts key financial details such as repayment duration, interest rates, and potential hidden fees. These details are automatically integrated into the financial dashboard's data model.

3. Behavioral Intervention Flow (The Nudge)

The most critical flow represents the notification triggered after a BNPL purchase. The prototype illustrates how Bijak Pay delivers an AI-generated message explaining the financial impact of the purchase in plain language, presenting users with a clear option to proceed or reconsider their decision.

Figma Prototype Development

A high-fidelity interactive prototype was created in Figma to visualize the complete user experience. The design emphasizes the readability of financial information by employing strategic color psychology—utilizing stark alert colors (like red) to highlight high-interest obligations and neutral tones for standard informational inputs. This prototype serves as a functional demonstration of Bijak Pay's capabilities during the project pitch.

4.0 Results

The primary outcome of this project is the development of a high-fidelity, interactive prototype for Bijak Pay, demonstrating a proposed solution for improving awareness and management of Buy Now, Pay Later (BNPL) financial commitments. Developed using Figma, the prototype illustrates the complete user journey from onboarding to ongoing debt monitoring, allowing the team to simulate how the system functions as a proof of concept. Through the prototyping process, the project successfully translated the conceptual solution into a visual and interactive interface that demonstrates how users can better understand and manage their BNPL obligations. The prototype focuses on three key functional outcomes aligned with the project objectives.

4.1 Centralized Financial Tracking

The prototype includes a centralized dashboard designed to aggregate multiple BNPL commitments into a single interface. Instead of requiring users to monitor separate BNPL platforms individually, the dashboard presents a unified overview of total outstanding debt, upcoming repayment schedules, and payment deadlines. By consolidating this information into a clear visual hierarchy, the interface reduces the cognitive effort required for users to understand their financial obligations.

4.2 Visualization of Behavioral Intervention

A key feature demonstrated in the prototype is the behavioral intervention mechanism, referred to as the "post-purchase nudge". This interaction flow simulates a notification triggered immediately after a BNPL purchase is detected. The interface displays a simplified breakdown of the financial impact of the purchase, including estimated total repayment costs over time. By presenting this information during the immediate post-purchase window, the system successfully visualizes a method to encourage users to reconsider impulsive purchases while refund options may still be available.

4.3 Simulated Automated Invoice Analysis

The prototype also demonstrates a simulated AI-assisted invoice analysis feature. In this interaction flow, users upload a BNPL receipt or invoice screenshot. The interface then illustrates how key financial details such as interest rates, repayment duration, and potential hidden fees can be automatically extracted and displayed in simplified terms. These extracted details are integrated into the central dashboard, providing users with a clearer understanding of their financial commitments.

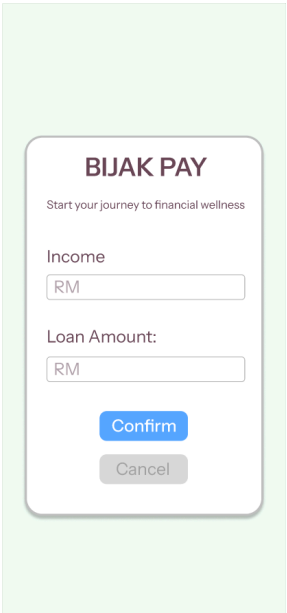
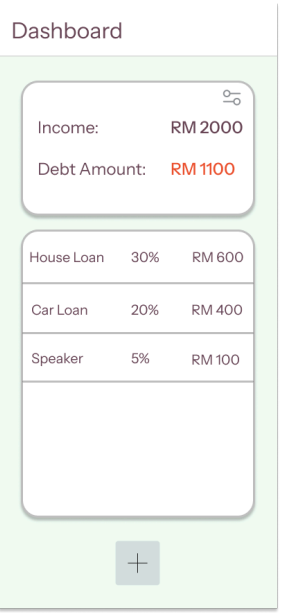
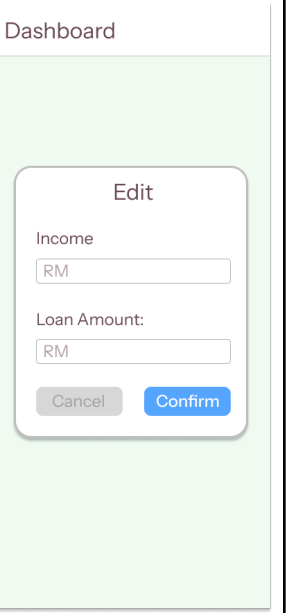
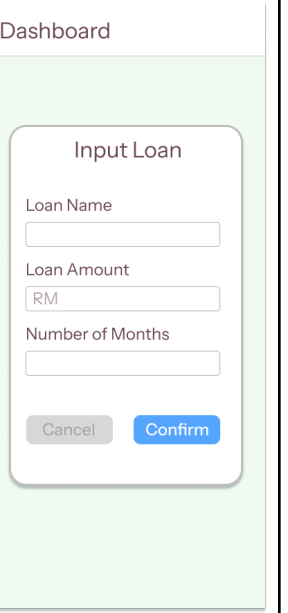
4.4 Proof of Concept Validation

Rather than conducting end-user testing, the evaluation of this project focused on validating the Proof of Concept (PoC) against the initial market research. The research phase confirmed that existing financial tools are largely reactive and do not address the behavioral psychology of impulse buying. The Bijak Pay prototype successfully demonstrates the feasibility of combining financial data visualization with proactive behavioral intervention techniques. By mapping out the specific notification triggers and UI flow, the prototype proves that intercepting a transaction at the invoice-generation stage is a viable user experience. The design illustrates how AI-assisted financial analysis and user-centered interface design can be integrated into a novel solution that fills the current gap in the market, promoting more responsible financial decision-making without being redundant to existing budgeting apps.

5.0 Prototype [Output & UI Screenshot]

Flow 1: Aggregated Dashboard Flow

This first flow demonstrates the application's centralized dashboard, designed to help users keep track of their ongoing debts, monthly payments, and financial commitments. Users begin by entering their monthly income and current loans, allowing the system to visually represent how much of their earnings are already claimed by existing debts. The interface provides a clear summary of total outstanding balances and allows users to easily add, edit, or monitor their monthly installments in one place.

 The screenshot shows the 'BIJAK PAY' onboarding screen. It has a light green background. At the top, it says 'BIJAK PAY' and 'Start your journey to financial wellness'. Below this are two input fields: 'Income' with a value of 'RM' and 'Loan Amount:' with a value of 'RM'. At the bottom are two buttons: 'Confirm' (blue) and 'Cancel' (grey).	 The screenshot shows the 'Dashboard' screen. It has a light green background. At the top, it says 'Dashboard'. Below this is a summary card showing 'Income: RM 2000' and 'Debt Amount: RM 1100'. Below the summary card is a table of loans: 'House Loan' (30%, RM 600), 'Car Loan' (20%, RM 400), and 'Speaker' (5%, RM 100). At the bottom is a '+' button.	 The screenshot shows the 'Edit' screen. It has a light green background. At the top, it says 'Edit'. Below this are two input fields: 'Income' with a value of 'RM' and 'Loan Amount:' with a value of 'RM'. At the bottom are two buttons: 'Cancel' (grey) and 'Confirm' (blue).	 The screenshot shows the 'Input Loan' screen. It has a light green background. At the top, it says 'Input Loan'. Below this are three input fields: 'Loan Name', 'Loan Amount' with a value of 'RM', and 'Number of Months'. At the bottom are two buttons: 'Cancel' (grey) and 'Confirm' (blue).
Screen 1 (Onboarding)	Screen 2 (Visualization)	Screen 3 (Dynamic Editing)	Screen 4 (Inputting New Debt)

Screen 1 (Onboarding): The user starts by entering their monthly income (money in) alongside their current active loans (money out). This establishes the baseline for their financial data.

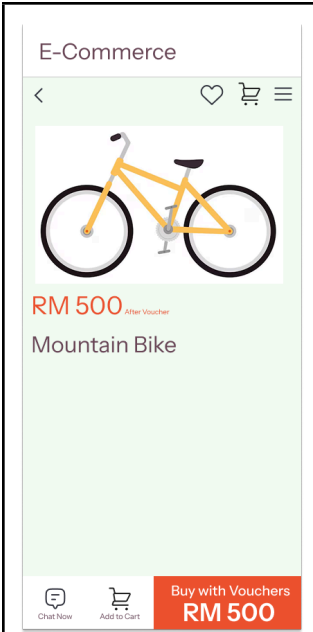
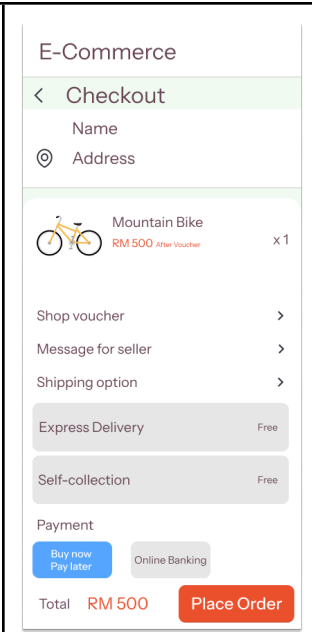
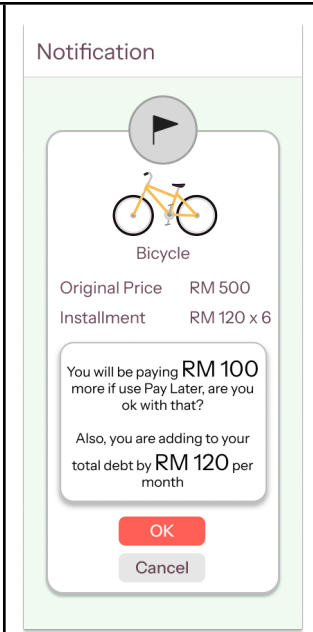
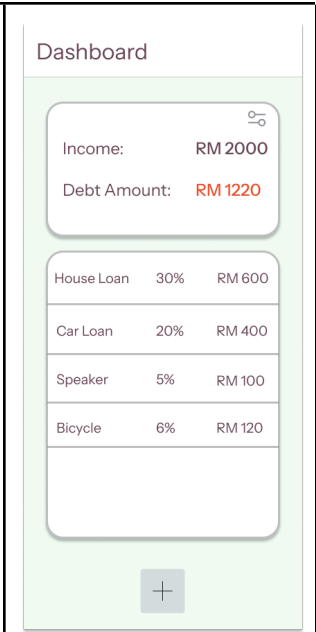
Screen 2 (Visualization): The dashboard screen assists the user in visualizing exactly how much of every cent earned is already claimed by existing loans before it even hits their pocket. The visual hierarchy emphasizes the total debt amount against the monthly income.

Screen 3 (Dynamic Editing): A dynamic function that allows the user to update and change their income or debt numbers if their financial situation changes, ensuring the dashboard remains an accurate reflection of their current obligations.

Screen 4 (Inputting New Debt): A straightforward form designed to add a new monthly payment by logging the loan name, total amount, and exactly how many months are left to pay.

Flow 2: E-Commerce Purchase and Behavioral Intervention Flow

This second flow illustrates the user journey when buying an item from an e-commerce platform using a "Buy Now, Pay Later" (BNPL) option. When a purchase is initiated and an invoice email is generated, Bijak Pay detects and automatically parses the document. The system then calculates the original price versus the true BNPL price, extracting any interest rates or hidden fees. It immediately delivers a proactive notification—the "nudge"—to explain the real financial impact of the installment plan, and subsequently updates the user's dashboard to reflect their newly reduced spending power

			
Screen 1 (The Trigger)	Screen 2 (The Illusion of Affordability)	Screen 3 (The Behavioral Nudge)	Screen 4 (Real-Time Impact)

Screen 1 (The Trigger): A standard e-commerce platform page where the user selects a product, such as a bicycle, to purchase.

Screen 2 (The Illusion of Affordability): The checkout screen presents "Buy Now, Pay Later" options, making an expensive item appear more affordable by breaking the total cost into small, monthly installments.

Screen 3 (The Behavioral Nudge): Just before the transaction is finalized, a notification pops up that teaches the user financial literacy in the moment. It explicitly explains how much interest they are losing and how this new debt will shrink their future monthly budget.

Screen 4 (Real-Time Impact): If the user chooses to proceed with the purchase, the Bijak Pay dashboard updates immediately. The new "Bicycle" debt is added to the list, clearly showing the user they now have less spending money available every month.

6.0 AI Acknowledgement

This project utilized several AI-assisted tools to support aspects of writing, design, and presentation preparation. These tools were used strictly as supplementary aids for editing, formatting, and content refinement, while all core ideas, analysis, and design decisions were developed by the project team.

Text Generation and Editing

1. **Gemini:** Used to refine the structural flow of the written report, improve grammatical accuracy, and assist in polishing sections of the presentation script.
2. **ChatGPT:** Utilized to assist with drafting evaluation rubrics, conducting preliminary self-assessment of report drafts, and providing suggestions for improving clarity and organization of the written content.

Visual and Multimedia Design

1. **Prezi AI:** Used to generate the initial layout and structural framework of the presentation slides.
2. **Canva AI:** Used to generate synthetic voice-over narration for multimedia presentation components.

All AI-generated suggestions and outputs were carefully reviewed, edited, and validated by the project team to ensure accuracy and alignment with the objectives of the project.

7.0 Conclusion

This project addresses the growing risk of "Buy Now, Pay Later" (BNPL) debt traps by shifting financial management from reactive budgeting to proactive behavioral intervention. By developing a comprehensive UI/UX prototype, the project demonstrates how AI can demystify complex financial terms and provide users with a transparent, centralized view of their total debt obligations. The proposed solution directly combats impulse purchasing through a timely, AI-driven "nudge" during the critical post-purchase window, empowering users to make informed decisions before committing to long-term debt. Ultimately, this prototype acts as a foundational step toward promoting greater financial literacy (SDG 4) and building economically resilient communities (SDG 11).

Potential Future Improvements While the current Figma prototype successfully visualizes the core intervention loop, a fully developed application could be expanded with the following features:

1. **Open Banking API Integration:** The application could securely sync with users' bank accounts and BNPL platforms to automatically pull real-time transaction data, eliminating the need for manual invoice uploads.
2. **Gamification of Financial Health:** Implementing a reward system—such as badges or credit score estimations—for consistently paying off debts early or maintaining a low BNPL utilization rate to reinforce positive financial habits.
3. **Bite-Sized Educational Modules:** Expanding on SDG 4 by integrating short, interactive lessons within the app to teach users about compounding interest, credit scores, and emergency funds.
4. **Predictive Financial Forecasting:** Leveraging AI to project a user's financial health 6 to 12 months into the future based on their current BNPL spending trajectory, moving beyond simple current debt tracking.

References

- Barroso, A., Mulka, L., McMullen, L., & de la Fuente, P. (2026, January 12). *How to Choose the Best Personal Finance Software and Apps*. NerdWallet.
<https://www.nerdwallet.com/article/finance/best-personal-finance-software>
- Feiam, A. (2025, June 9). *Big changes coming to BNPL from Tuesday*.
<https://www.news.com.au/finance/money/big-changes-are-coming-to-buy-now-pay-later-heres-how-it-affects-you/news-story/866ab968cd5f1e77ce13dd30d9f5847f>
- Financial Conduct Authority. (2025, July 16). *Regulating Buy Now Pay Later (BNPL)*. FCA.
<https://www.fca.org.uk/firms/regulating-buy-now-pay-later>
- Financial Conduct Authority. (2026, February 10). *New protections confirmed for Buy Now Pay Later borrowers*. FCA.
<https://www.fca.org.uk/news/press-releases/new-protections-confirmed-buy-now-pay-later-borrowers>
- Firefly III. (2023, April 22). *Firefly III*. GitHub. <https://github.com/firefly-iii/firefly-iii>
- Greenberg Traurig LLP. (2024). *CFPB Issues Interpretive Rule Classifying BNPL Products as “Credit Cards”* | Insights | Greenberg Traurig LLP. Gtlaw.com.
<https://www.gtlaw.com/en/insights/2024/6/cfpb-issues-interpretive-rule-classifying-bnpl-products-as-credit-cards>
- Maggio, M. D., Williams, E., & Katz, J. (2022, September 1). *Buy Now, Pay Later Credit: User Characteristics and Effects on Spending Patterns*. National Bureau of Economic Research. <https://www.nber.org/papers/w30508>
- Money Manager Ex. (2026, January 20). *GitHub - moneymanagerex/moneymanagerex: Money Manager Ex is an easy to use, money management application built with wxWidgets*. GitHub. <https://github.com/moneymanagerex/moneymanagerex>
- PersonalFinances. (2026, February 22). *Best Budgeting Apps in 2026: A Complete Comparison*. Personal Finances.
<https://personalfinances.com/blog/budgeting/budgeting-apps-comparison/>
- Shah, A. (2026). *Study of Buy-Now-Pay-Later (BNPL) Adoption Among Students and Its Impact on Spending Patterns*. Rjwave.org.
<https://rjwave.org/jaafr/viewpaperforall.php?paper=JAAFR2512458>